

Assignment 6: Pseudocode for Emergency Department at-risk rule

Introduction

In the Emergency Department triage, at-risk rules are criteria used to identify patients who are vulnerable, have a high risk of rapid deterioration or require urgent medical attention. These rules are based on physiological observations such as respiratory rate, oxygen saturation (SpO₂), blood pressure, heart rate, glasgow coma scale (GCS) and temperature. Through detecting abnormal findings early, clinicians can quickly prioritise critically ill patients, improve patient safety and ensure patients who require immediate care receive timely treatment.

Physiological ranges

Indicators	At - Risk Range
Oxygen Saturation (SpO ₂)	< 92%
Respiratory Rate (RR)	< 12 OR > 20 - 25 breaths/min
Blood Pressure (BP)	Systolic 130-139 mmHg OR Diastolic 80-89 mmHg
Heart Rate (Bpm)	< 60 OR > 100 bpm
Temperature (°C)	< 35 OR > 40
Glasgow Coma Scale (GCS)	< 8

Table 1. Physiological Ranges for At-Risk Patients in ED

ED Triage Pseudocode

START TRIAGE

Collect:

```
SpO2
respiratory_rate
systolic_BP
Diastolic_BP
heart_rate
temp_c
GCS
```

```

SET triage_cat = "Standard"

#-----
# Temp Conversion
#-----

temp_f = (temp_c * 9/5) + 32

#-----
# Respiratory
#-----

IF SpO2 < 92 THEN
    triage_cat = "Critical"
END IF

IF respiratory_rate < 12 or respiratory_rate > 25 THEN
    triage_cat = "High Priority"
END IF

#-----
# Circulation
#-----

IF systolic_BP < 130 or systolic_BP > 139
OR Diastolic_BP < 80 or Diastolic_BP > 89 THEN
    triage_cat = "Elevated BP (monitor)"
END IF

IF heart_rate < 60 or heart_rate > 100 THEN
    triage_cat = "High Priority"
END IF

#-----
# Temperature
#-----

IF temp_c < 35 OR temp_c > 40 THEN
    triage_cat = "Critical"
END IF

```

```

IF temp_f < 95 OR temp_f > 104 THEN
    triage_cat = "Critical"
END IF

#-----
# GCS
#-----

IF GCS < 8 THEN
    triage_cat = "Critical"
END IF

#-----
# Final assessment
#-----

IF triage_cat = "Critical" THEN
    Patients receive medical attention

ELSE IF triage_cat = "High Priority" THEN
    Assign urgent review

ELSE
    Continue standard triage assessment

END IF

END TRIAGE

```

Explanation

The above pseudocode shows how the ED triage system can use vital signs to quickly assess a patient's severity. It collects standard ranges for SpO₂, respiratory rate, blood pressure, heart rate, temperature (°C and °F) and GCS, then compares them to at-risk thresholds.

Abnormal values are flagged and temperature is also converted from celsius to fahrenheit for consistency across various systems. Finally, a final assessment is used to assign patients to critical, high priority or standard care based on how many risk thresholds are triggered. This ensures that the most unstable patients are treated first.